

Technical Report: ZPSR vs. Unconstrained BPE

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Abstract

This technical report formalizes the comparative proof that grammar-constrained token generation (Zero-Shot Polysynthetic State Resolver, ZPSR) produces fewer, more meaningful tokens than statistical Byte Pair Encoding (BPE) when applied to polysynthetic languages.

Primary Innovation: Replace Shannon entropy (transcendental $\ln(x)$, measures disorder) with anti-Shannon purity ($N \times \text{IPR}$, hardware-native polynomial, measures information localization).

Status: Benchmark MEASURED 2026-08-30 (tiktoken 0.12.0; ALTLab crk FSTs via hfst-optimized-lookup 0.0.14). Hypothesis verdicts withdrawn pending replication on ALTLab corpus.

Erratum (v2): The v1 figures (13,000 vs. 5,000 tokens; 42.3% vs. 100% validity; $N \times \text{IPR}$ 1.2 vs. 195.0; 1.81 vs. 4.71 B/token) were constants written into `run-zpsr-vs-bpe-v2.ps1`, not measurements. They are withdrawn. Every number in this version is computed from the corpus by `.forge/benchmark/measure_zpsr_vs_bpe.py`; the procedure is in `METHOD.md`.

1 Executive Summary

Four hypotheses; measured values below, verdicts pending replication on ALTLab corpus. BPE = gpt2 / cl100k_base (tiktoken). ZPSR = inflectional-boundary segments from the ALTLab strict analyser + boundary generator (see `METHOD.md`). Corpus: 2,443 whitespace words, 28,662 bytes.

Hypothesis	Metric	BPE (gpt2 / cl100k)	ZPSR
H1	Token count, same 1,587 analysed words	8,363 / 7,393	2,545
H2	FST coverage (validity on corpus)	n/a (no oracle)	64.96% strict / 74.62% relaxed
H3	Effective vocab $1/\Sigma p_i^2$	39.9 / 52.2	89.0
H4	Bytes/token, same 15,932 bytes	1.905 / 2.155	6.260

2 Measurement Framework

2.1 Classical Entropy vs. Anti-Shannon Purity

Aspect	Shannon H_1	N×IPR (Anti-Shannon)
Formula	$-\sum(p_i \ln(p_i))$	$N \times \sum(p_i^2)$
Meaning	Disorder/diffusion	Purity/localization
Hardware	Transcendental ln (20–30 cycles)	FMA dot-product (2 cycles)
BPE context	High H_1 (chaos)	Low N×IPR (scattered)
ZPSR context	Low H_1 (order)	High N×IPR (concentrated)
Intuition	“Lower entropy is better”	“Higher purity is better”

Hardware Advantage of N×IPR:

Shannon H_1 latency per token:	20--30 cycles (transcendental stall)
N×IPR latency per token:	2 cycles (FMA, no stalls)
Speedup:	10--15×
Cache footprint:	L1 shared memory (256 bytes fits 64 tokens)

2.2 Why BPE Fails Polysynthetic Languages

BPE is trained on **English-dominant corpora**. When applied to Cree:

- Out-of-Vocabulary (OOV) Characters:** UCAS syllabics (U+1400–U+167F) absent from training data
 - Result: Fallback to UTF-8 byte decomposition
 - Cost: 1–3 tokens per UCAS character
- Morpheme Destruction:** BPE merges frequent byte-pairs, not morpheme boundaries
 - Cree word: “wâpamêw” — strict analyser: wâpamêw+V+TA+Ind+3Sg+4Sg/P10; inflectional segmentation wâpam>êw (2 segments). Morpheme gloss: verify with a speaker.
 - BPE tokenization (measured): gpt2 and cl100k both give w | â | p | am | ê | w = 6 tokens
 - ZPSR tokenization (measured): 2 segments
- No Grammar Enforcement:** BPE treats all tokens as equally probable
 - Missing mandatory constraints (animacy, obviation, direction hierarchy)
 - Result: Possible but ungrammatical output passes through

3 Experimental Design

3.1 Corpus Specification

Parameter	Value
Language	Plains Cree (nêhiyawêwin)
Size	2,443 whitespace words
Source	GiellaLT lang-crk devtools/testdata/crk_public_texts.txt (upstream commit 63)
Verification	None. Unverified test file; contains English words, URLs and phone numbers
Total bytes (UTF-8)	28,662 bytes
Hash	SHA-256 c8fa6e740bdb9d76efaedc125efdcc941f3b3f8ff52a6fc5ed4aa483e08e9c9c

3.2 Tokenization Procedures

BPE (Baseline):

```

Input: Plains Cree text
Vocab: tiktoken 0.12.0, gpt2 (N=50,257) and cl100k_base (N=100,277)
Process:
  1. enc.encode(whole corpus text)
  2. Count tokens: count_bpe
  3. Calculate distribution: p_i = count(token_i) / count_bpe
  4. Compute  $\Sigma(p_i^2)$ ,  $N \times \text{IPR}$ , effective vocab  $1/\Sigma(p_i^2)$ 
  5. Validity: NOT computed -- no oracle maps a byte-piece to a Cree morpheme (null)
Output: (count_bpe,  $\Sigma p^2$ ,  $N \times \text{IPR}$ ,  $1/\Sigma p^2$ )

```

ZPSR (Grammar-Constrained):

```

Input: Plains Cree text
FST:   ALTLab morphodict crk-strict-analyzer-for-dictionary.hfstol
       + crk-strict-generator-with-morpheme-boundaries.hfstol (main @ e3468e1)
Process (per whitespace word; GBNF/ASP layers are NOT part of this measurement):
  1. Strip leading/trailing punctuation; strict analyser lookup (lowercase retry)
  2. No analysis -> unanalysed (0 tokens, counted separately)
  3. First analysis -> boundary generator -> surface form with '<' '>' marks
  4. Split on boundary marks -> ZPSR tokens (inflectional segments)
  5. Count tokens: count_zpsr; distribution p_i over segment types
  6. Compute  $\Sigma(p_i^2)$ ,  $1/\Sigma(p_i^2)$ ;  $N \times \text{IPR}$  with N = observed types (no fixed vocab)
  7. Coverage = analysed words / all words (this IS the validity figure)
Output: (count_zpsr,  $\Sigma p^2$ ,  $1/\Sigma p^2$ , coverage)

```

3.3 Measured Metrics

Metric	Definition	Unit	Observable
Token Count	Total tokens after encoding	tokens	<code>len(tokenize(text))</code>
$N \times \text{IPR}$ Purity	Concentration on valid states	scalar $\in [1, N]$	$N \times \Sigma(p_i^2)$
Coverage %	Words the strict analyser accepts	%	analysed / all whitespace words (B)
Compression Ratio	Bytes per token	B/token	<code>raw_bytes / token_count</code>
Delta Metrics	Comparison BPE vs. ZPSR	% change	Relative improvement

4 Hypotheses and Predictions

4.1 H1: Grammar Reduces Token Count

Null: $\text{count}_{\text{zpsr}} \approx \text{count}_{\text{bpe}}$ (no difference) **Alternative:** $\text{count}_{\text{zpsr}} < \text{count}_{\text{bpe}}$ (ZPSR uses fewer tokens)

Rationale:

- Polysynthetic morphology: $\sim 4\text{--}5$ morphemes per word
- BPE: statistically arbitrary merging $\rightarrow \sim 5\text{--}7$ tokens per word
- ZPSR: morpheme-aligned tokenization $\rightarrow \sim 1\text{--}2$ tokens per morpheme cluster
- Prediction: **15–40% token savings** (conservative: 20% expected)

Success Criterion: $\text{count}_{\text{zpsr}} < \text{count}_{\text{bpe}}$ by $> 10\%$ ($p < 0.05$)

Result: measured -69.6% (gpt2) / -65.6% (cl100k) on the 1,587 analysed words (2,545 vs. 8,363 / 7,393). No significance test was run. Verdict: [pending replication on ALTLab corpus].

4.2 H2: FST Guarantees Linguistic Validity

Null: $\text{validity}_{\text{zpsr}} \approx \text{validity}_{\text{bpe}}$ **Alternative:** $\text{validity}_{\text{zpsr}} > \text{validity}_{\text{bpe}}$

Rationale:

- BPE: validity not computable — no oracle maps a byte-piece to a Cree morpheme (reported as null)
- ZPSR: validity on a corpus is analyser coverage; measured 64.96% strict, 74.62% relaxed
- Prediction withdrawn; [pending replication on ALTLab corpus]

Success Criterion: $\text{validity}_{\text{zpsr}} \geq 95\%$, $\text{validity}_{\text{bpe}} < 70\%$

Result: coverage 64.96% strict (1,587/2,443) / 74.62% relaxed (1,823/2,443); BPE validity null. 856 words unanalysed (English, URLs, numbers, and Cree forms absent from the lexicon).

4.3 H3: Grammar Creates Pure Token States ($N \times \text{IPR}$)

Null: $\text{IPR}_{\text{zpsr}} \approx \text{IPR}_{\text{bpe}}$ **Alternative:** $\text{IPR}_{\text{zpsr}} > \text{IPR}_{\text{bpe}}$

Rationale:

- BPE: 50K vocabulary, scattered probability (near-uniform chaos)
 - $N \times \text{IPR} \approx 1$ (thermal spread across vocab)
- ZPSR: $\sim 100\text{--}200$ valid morpheme paths (discrete, finite set)
 - $N \times \text{IPR} \approx N/\sqrt{\text{paths}}$ (concentrated on valid subspace)
- Prediction: **IPR_{zpsr} 30–50% higher** ($10\text{--}13\times$ expected)

Success Criterion: $\text{IPR}_{\text{zpsr}} > \text{IPR}_{\text{bpe}}$ by $> 20\%$ ($p < 0.05$)

Result: $N \times \text{IPR}$ gpt2 1,260.2 / cl100k 1,922.2 / ZPSR 5.71 ($N=508$ observed types; 564.9 and 1,127.2 using the BPE N). Effective vocabulary $1/\Sigma p_i^2$: 39.9 / 52.2 / 89.0. The hypothesis as stated (BPE $N \times \text{IPR} \approx 1$) is not supported: Σp_i^2 for gpt2 is 0.02508, so $N \Sigma p_i^2 = 1,260$. [pending replication on ALTLab corpus]

4.4 H4: Better Compression Efficiency

Null: $\text{ratio}_{\text{zpsr}} \approx \text{ratio}_{\text{bpe}}$ **Alternative:** $\text{ratio}_{\text{zpsr}} > \text{ratio}_{\text{bpe}}$

Rationale:

- Same input bytes, fewer output tokens
- Bytes-per-token ratio improves (denominator shrinks more than numerator)
- Prediction: **20–40% compression improvement**

Success Criterion: $\text{compression_ratio}_{\text{zpsr}} / \text{compression_ratio}_{\text{bpe}} \geq 1.15$

Result: on the same 15,932 bytes (1,587 analysed words + spaces): gpt2 15,932/8,363 = 1.905 B/token; cl100k 2.155; ZPSR 15,932/2,545 = 6.260 B/token ($3.29\times$ / $2.91\times$). Whole-corpus BPE: 1.887 / 2.165 B/token.

5 Dual-Oracle Verification (C11)

5.1 Claim: “ZPSR produces fewer, more meaningful tokens than BPE”

Oracle 1 (Forbidden by Constraint) — Theoretical proof:

1. Polysynthetic morphology packages multiple morphemes per word
2. FST enforces only valid morpheme sequences (bounded set, deterministic)
3. BPE merges are statistically frequent byte pairs (unbounded, chaotic)
4. **Mathematical consequence:** $\text{Token count}(\text{ZPSR}) < \text{Token count}(\text{BPE})$
5. **Evidence:** Linguistic theory (Baker, 2001) + FST properties + agent findings (A1–A4)

Oracle 2 (Empirical Verification) — Benchmark measurement (August 30, 2026):

1. Tokenized the full 2,443-word lang-crkr test file (28,662 bytes UTF-8)
2. Measured token counts, Σp_i^2 , N×IPR, effective vocabulary, analyser coverage
3. **Results:** H1 and H4 directionally supported on the 65% of words the analyser covers; H2 as stated is false (coverage 64.96%, not 100%); H3 as stated is false (BPE N×IPR is in the thousands)
4. **Status:** MEASURED (script: `measure_zpsr_vs_bpe.py`; results: `zpsr-vs-bpe-results.json`; method: `METHOD.md`; receipts: `RECEIPTS.txt`)

Verdict: [pending replication on ALTLab corpus]

6 Empirical Results (Measured August 30, 2026)

6.1 Benchmark Output (excerpt of zpsr-vs-bpe-results.json)

```

''provenance'': {
  ''corpus_sha256'': ''c8fa6e740bdb9d76efaedc125efdcc941f3b3f8ff52a6fc5ed4aa483e08e9c9c''
  ,
  ''corpus_bytes'': 28662, ''corpus_whitespace_words'': 2443,
  ''lang_crk_git_head'': ''0c55b17941563277e32894f5fa1b99fff06380f4'',
  ''tiktoken_version'': ''0.12.0'', ''hfst_optimized_lookup_version'': ''0.0.14''
},
''bpe'': {
  ''gpt2'': { ''vocab_N'': 50257, ''token_count'': 15188, ''distinct_tokens'': 430,
    ''tokens_per_word'': 6.217, ''bytes_per_token'': 1.887,
    ''sum_p2'': 0.025076, ''N_ipr'': 1260.2, ''effective_vocab'': 39.9,
    ''on_fst_analysed_words_only'': { ''words'': 1587, ''bytes'': 15932, ''
      token_count'': 8363 } },
  ''cl100k_base'': { ''vocab_N'': 100277, ''token_count'': 13239, ''distinct_tokens'':
    537,
    ''tokens_per_word'': 5.419, ''bytes_per_token'': 2.165,
    ''sum_p2'': 0.019169, ''N_ipr'': 1922.2, ''effective_vocab'': 52.2,
    ''on_fst_analysed_words_only'': { ''words'': 1587, ''bytes'': 15932, ''
      token_count'': 7393 } }
},
''byte_level'': { ''vocab_N'': 256, ''token_count'': 28662, ''tokens_per_word'': 11.732,
  ''sum_p2'': 0.064143, ''effective_vocab'': 15.6 },
''zpsr'': {
  ''coverage'': { ''total_whitespace_words'': 2443, ''analysed_words_strict'': 1587,
    ''coverage_strict'': 0.6496, ''coverage_relaxed'': 0.7462,
    ''validity_100pct_claim'': false },
  ''morpheme_stream'': { ''token_count'': 2545, ''distinct_types'': 508,
    ''sum_p2'': 0.011241, ''effective_vocab'': 89.0,
    ''N_ipr_N_equals_distinct_types_observed'': 5.71,
    ''tokens_per_word_over_segmented_words'': 1.604,
    ''bytes_per_token_analysed_words_only'': 5.637 },
  ''tag_stream'': { ''tags_not_morphemes'': true, ''token_count'': 4191, ''distinct_types
    '': 56 }
}

```

6.2 Measured Effects (no statistical test was run)

Hypothesis	Observed Effect	Criterion (v1)	Result
H1	−69.6% / −65.6% tokens (1,587 analysed words)	> 10% reduction	measured; [p]
H2	coverage 64.96% strict; BPE null	≥ 95% vs. < 70%	not met; 100%
H3	1/Σp ² : 89.0 vs. 39.9 / 52.2; N×IPR 5.71 vs. 1,260 / 1,922	> 20% N×IPR increase	not supported
H4	3.29× / 2.91× bytes/token (same bytes)	≥ 1.15×	measured; [p]

7 Success/Failure Criteria

7.1 Success (All Required)

- **H1:** $\text{count}_{\text{zpsr}} < \text{count}_{\text{bpe}}$ by $> 10\%$ — measured -69.6% / -65.6% on analysed words; no p -value computed
- **H2:** $\text{validity}_{\text{zpsr}} \geq 95\%$ — NOT met (coverage 64.96%); $\text{validity}_{\text{bpe}}$ not computable
- **H3:** $\text{IPR}_{\text{zpsr}} > \text{IPR}_{\text{bpe}}$ by $> 20\%$ — NOT supported as stated; effective vocabulary 89.0 vs. 39.9 / 52.2
- **H4:** bytes/token ratio ≥ 1.15 — measured $3.29\times$ / $2.91\times$ on the same bytes
- **Dual-Oracle 1:** theoretical argument unchanged
- **Dual-Oracle 2:** [pending replication on ALTLab corpus]

8 References

- Morin, S. (2026). Sovereign Edge-Native Language Processing: ZPSR Whitepaper v4. *Zenodo*. DOI: 10.5281/zenodo.22020676.
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- Giellatekno. (2024). Plains Cree Finite-State Transducer. <https://github.com/giellalt/lang-crk>
- Baker, M. C. (2001). *The Atoms of Language*. Oxford University Press.
- Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*.

Status: Benchmark MEASURED 2026-08-30. Hypothesis verdicts [pending replication on ALTLab corpus]. v1 numbers withdrawn (see Erratum).

Owner: Sean Morin, 13forge

Date: August 30, 2026 (v2)

Benchmark artifact: `.forge/benchmark/zpsr-vs-bpe-results.json` + `RECEIPTS.txt`

Benchmark script: `.forge/benchmark/measure_zpsr_vs_bpe.py` (method: `METHOD.md`; v1 script kept as `run-zpsr-vs-bpe-v2.ps1.ESTIMATE-ONLY`)

Corpus: GiellaLT `lang-crk crk_public_texts.txt` (2,443 words, 28,662 bytes, unverified test file)